

How the fracture connection factors produce crossflow in Flow: a synthetic three-completion example

OPM Flow geomechanics – fracture coupling notes

October 1, 2026

Abstract

A small, open, circular fracture is placed on a vertical injector with three completions in an (almost) hydrostatic reservoir, and injection is at a fairly low rate. We follow, step by step, what the fracture model in `opm-flowgeomechanics` computes (fracture pressure, width, leak-off), how `Fracture::wellIndices()` turns the leak-off into connection transmissibility factors (CTFs), and how Flow’s standard well model then applies these factors. The fracture model predicts injection into all three layers. Flow, using the same factors, injects into the top completion and *produces* from the bottom completion as soon as the potential that Flow assigns to a connection differs from the potential the CTF was derived for by a depth-dependent amount s_k . We derive the exact condition: crossflow occurs whenever the rate is below $Q_{crit} = \sum_k (T_k + M_k)(s_k - s_{min})$. The fracture CTFs are large because they represent leak-off over the whole fracture face, while at low rates the overpressure per connection is a few hundredths of a bar. So a mismatch of a few hundredths of a bar is enough to reverse the deepest connection. We list the sources of s_k in the code, show the singular behaviour of the legacy CTF formula when layer potentials differ, follow the sequential coupling in time, and evaluate the two remedies `wi_flow_column` (option 2) and `wi_fracture_pressure` (option 3). Up to Section 9 the default `wi_upscaling = legacy` is used; that section repeats the analysis for `wi_upscaling = conductivity`. Without normalisation that mode crossflows more than legacy. With the default normalisation and pressure floor it crossflows less, but only because at low rates it scales the fracture connections down, so that Flow injects just a fraction of the modelled leak-off through the fracture. Section 10 repeats the analysis with the properties of `test/m51w/TEST.DATA` and adds formation damage (`WINJDAM`). With the deck’s 1000 mD, the overpressure per connection is only 0.03 bar, but the layers re-equilibrate quickly, so sustained crossflow needs a mismatch of 20–30 % of the hydrostatic gradient. Formation damage raises that threshold further, but its cake is deposited where the *fracture model* sees leak-off, not where Flow injects, which can reinforce an existing skew.

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1 Set-up

1.1 Geometry and properties

The reservoir consists of three horizontal layers of thickness $h = 10 \text{ m}$, discretised by one column of cells of size $\Delta x = \Delta y = 20 \text{ m}$ (Figure 1). A vertical water injector is completed in all three cells, at the cell-centre depths $z_1 = 1990$, $z_2 = 2000$, $z_3 = 2010 \text{ m}$ (depth positive downwards). A vertical, circular fracture of radius $R = 12 \text{ m}$ is centred on the middle completion (seed depth $z_o = 2000 \text{ m}$). It reaches into the top and bottom layers but not beyond them.

quantity	symbol	value
permeability (isotropic)	k	100 mD
water viscosity, mobility	$\mu, \lambda = 1/\mu$	0.5 cP, 2000 (Pa s) ⁻¹
water density (reservoir and fracture model)	ρ_r	1020 kg/m ³
leak-off distance (legacy: $0.25\Delta x$)	d	5 m
leaking fracture faces (legacy)	n_s	2
Young's modulus, Poisson's ratio	E, ν	10 GPa, 0.25
net pressure at the seed	p_{net}	2.0 bar
wellbore radius, Peaceman radius	$r_w, r_o = 0.198 \Delta x$	0.1 m, 3.96 m
well $\rightarrow$ fracture conductance (ring feed)	W	$100 \sum_k L_k$
injection rate (reservoir volumes)	Q	8 m ³ /d

Table 1: Parameters of the example. All rates are reservoir volumes per day, all potentials are in bar, and all conductances (including the mobility) are in m³/(d bar).

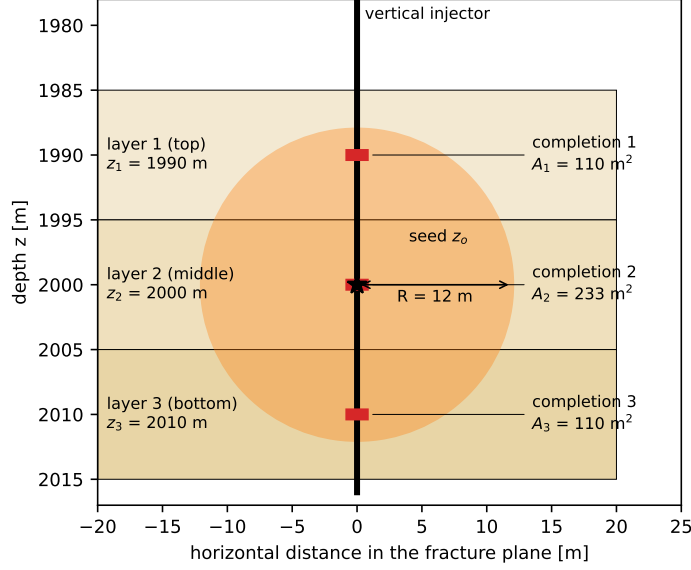


Figure 1: Side view in the fracture plane. Three layers, one completion per layer, a circular fracture of radius 12 m centred on the middle completion. A_k is the fracture area inside layer k .

1.2 Potentials

Everything below is written in terms of the potential

$$\Phi = p - p_o - \rho_r g (z - z_o) \quad [\text{bar}], \quad (1)$$

where p_o is the reservoir pressure at the seed depth. This is exactly the quantity the fracture code uses: `fracture_dgh = g\rho z` is subtracted from the fracture pressure, and $g\rho z_r$ from the reservoir pressure (`Fracture::leakOffRate()`). The reservoir is hydrostatic, so the three layer potentials are equal, and we set them to zero: $\Phi_1 = \Phi_2 = \Phi_3 = 0$. The well potential Φ_w is the potential of the injection pressure at the seed, $\Phi_w = p_{inj} - p_o$, which is what the code calls `inj_press - dh_perf`.

Why the overpressures are small. An open fracture needs $p_f > \sigma_h$. This does *not* mean that the fracture is far above the pressure of the cells it cuts. With continued injection, the cells around the fracture are pressurised (and the total stress rises with them through poroelastic back-stress), while the fracture-to-cell connection has a very large conductance (a large face area over a short distance, $d = \Delta x/4$). The quantity that matters for the connections is the fracture-to-cell potential difference. At a rate Q it is of order $Q/\sum_k L_k$, which for the numbers here is a few hundredths of a bar. We prescribe the open state through the net pressure 2.0 bar that enters the width (Section 2) and do not model how it was reached.

2 The fracture model

2.1 Leak-off conductances

The area of the disk between depths a and b (relative to the centre) is

$$A(a, b) = \left[z\sqrt{R^2 - z^2} + R^2 \arcsin(z/R) \right]_{z=\max(a, -R)}^{z=\min(b, R)}, \quad (2)$$

which gives $A_1 = A_3 = 109.8 \text{ m}^2$ and $A_2 = 232.9 \text{ m}^2$ (total $\pi R^2 = 452.4 \text{ m}^2$). The leak-off model of the code (`Fracture::updateLeakoff()`, legacy) is

$$\ell_i = n_s \frac{\lambda k \Delta A_i}{d} \quad \implies \quad L_k = \sum_{i \in k} \ell_i = n_s \frac{\lambda k A_k}{d}, \quad (3)$$

for every fracture cell i with area ΔA_i , and per layer k . This gives

$$L_1 = L_3 = 74.9, \quad L_2 = 158.8, \quad \sum_k L_k = 308.6 \text{ m}^3/(\text{d bar}).$$

For comparison, the matrix CTF of each completion (Peaceman, with mobility) is

$$M_k = \frac{2\pi k h \lambda}{\ln(r_o/r_w)} = 29.1 \text{ m}^3/(\text{d bar}). \quad (4)$$

The fracture adds 2.6 to 5.5 times the matrix connectivity to each completion.

2.2 Width, fracture flow and leak-off

The width is that of a penny-shaped crack under uniform net pressure (Sneddon),

$$w(r) = \frac{8(1-\nu^2)}{\pi E} p_{net} \sqrt{R^2 - r^2}, \quad w_{\max} = 0.57 \text{ mm}. \quad (5)$$

The fracture plane is discretised by 96×96 square cells (side 0.25 m), of which those inside the disk are active. For each active cell i with potential Φ_i (here $\Phi_i = p_i - p_o - \rho_r g(z_i - z_o)$, the fracture's own gravity term), the steady mass balance is

$$\sum_{j \in \mathcal{N}(i)} T_{ij} (\Phi_i - \Phi_j) + \ell_i (\Phi_i - \Phi_{k(i)}) = w_i (\Phi_w - \Phi_i), \quad T_{ij} = \frac{\lambda}{12/w_i^3 + 12/w_j^3}, \quad (6)$$

with the cubic-law face transmissibility T_{ij} . Here $k(i)$ is the layer holding the cell centre, and w_i (non-zero only on the ring of cells at $r < 0.6$ m) is the well feed, with $\sum_i w_i = W$. This is `Fracture::assemblePressure()` and `addSource()` with `perf_pressure` control, written in potentials. The leak-off into layer k is

$$q_k = \sum_{i \in k} \ell_i (\Phi_i - \Phi_k) = L_k (\bar{\Phi}_{F,k} - \Phi_k), \quad (7)$$

where $\bar{\Phi}_{F,k}$ is the leak-off-weighted mean fracture potential over the part of the fracture in layer k . Because (6) is linear, q_k is a linear function of Φ_w and of the layer potentials Φ_k . We compute this response once and use it throughout.

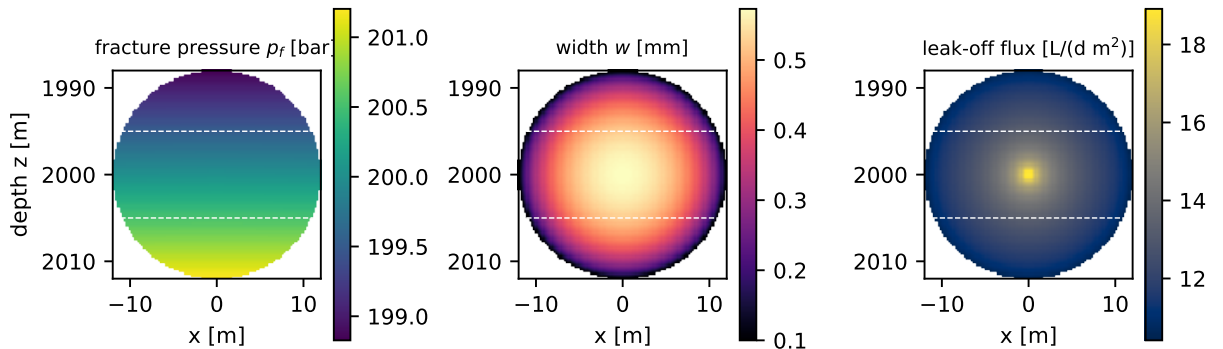


Figure 2: Fracture model at $Q = 8 \text{ m}^3/\text{d}$: pressure (hydrostatic gradient, top-down), width (symmetric), leak-off flux per unit area (dashed lines: layer boundaries). The leak-off is positive everywhere, and its mild radial variation comes from the viscous pressure drop away from the feed ring.

2.3 Reference state at $Q = 8 \text{ m}^3/\text{d}$

The fracture is fed from the well, and the matrix completions inject directly. The well potential that delivers the rate Q is found from

$$\sum_k q_k(\Phi_w) + \sum_k M_k(\Phi_w - \Phi_k) = Q. \quad (8)$$

The result is $\Phi_w = 0.0278 \text{ bar}$, with a mean fracture potential $\bar{\Phi}_F = 0.0181 \text{ bar}$. The difference of 0.0097 bar is the pressure drop in the feed ring and in the fracture. The leak-off and matrix rates are

$$q = (1.28, 3.01, 1.28), \quad M(\Phi_w - \Phi) = (0.81, 0.81, 0.81) \text{ m}^3/\text{d}.$$

All three layers receive water from the fracture. This is what the fracture model predicts, and it is also what a fully coupled model (fracture cells as unknowns of the flow problem) would compute. It is the reference in the following.

3 From leak-off to connection factors (the code)

`Fracture::wellIndices()` collects the leak-off per reservoir cell and divides it by the well-to-cell potential difference. This is the default, `wi_upscaling = legacy`, which is assumed up to Section 9:

```
double dh_res = z_cells[i] * gravity_ * dens_cells[i]; // rho_r g z_k
double dh_perf = gravity_ * perf_density * origo_[2]; // rho_r g z_o
double dp_well = (inj_press - dh_perf) - (p_cells[i] - dh_res);
WI = q_cells[i] / dp_well; // q_cells: leak-off summed per cell
ctf = WI / mob_cells[i]; // negative WI -> 0
```

In potentials this reads

$$T_k^{\text{leg}} = \lambda \text{CTF}_k = \frac{q_k}{\Phi_w - \Phi_k} = L_k \frac{\bar{\Phi}_{F,k} - \Phi_k}{\Phi_w - \Phi_k}, \quad T_k^{\text{leg}} \leftarrow 0 \text{ if negative.} \quad (9)$$

At the reference state

$$T^{\text{leg}} = (46.2, 108.4, 46.2) \text{ m}^3/(\text{d bar}), \quad T_k^{\text{leg}}/L_k = (0.617, 0.683, 0.617).$$

These CTFs are passed to Flow (`addFracturePerforations`) and *added* to the matrix CTF of the existing completion in `WellInterface::getTw()`.

4 What Flow does with these factors

4.1 Flow's connection model

For a standard well, Flow computes the flux of connection k as

$$q_k^F = (T_k + M_k) (p_{bhp} + \Delta p_k^{wb} - p_{cell,k}), \quad (10)$$

where Δp_k^{wb} is Flow's hydrostatic wellbore term (computed in `StandardWellConnections`) and $p_{cell,k}$ is the cell pressure used by the well model (oil pressure if oil is active, else gas, else water; see `getPerfCellPressure`). Written with the potential (1) and referred to the seed,

$$q_k^F = (T_k + M_k) (\Phi_w^F + s_k - \Phi_k) \quad (11)$$

where s_k collects *everything by which Flow's connection potential differs from the potential Φ_w that the CTF formula (9) assumed*:

$$s_k = \underbrace{(p_{bhp} + \Delta p_k^{wb}) - (p_{bhp} + \Delta p_o^{wb}) - \rho_r g(z_k - z_o)}_{\text{wellbore column vs. } \rho_r} - \underbrace{(p_{cell,k} - p_{w,k})}_{\text{cell pressure vs. water pressure}} + \dots \quad (12)$$

Under rate control, Flow chooses Φ_w^F so that $\sum_k q_k^F = Q$:

$$\Phi_w^F = \frac{Q - \sum_k (T_k + M_k)(s_k - \Phi_k)}{\sum_k (T_k + M_k)}. \quad (13)$$

4.2 No mismatch: Flow reproduces the fracture model

With $s_k = 0$ and $\Phi_k = 0$, (13) gives $\Phi_w^F = Q / \sum_k (T_k + M_k)$. By construction of (9), $\sum_k T_k \Phi_w = \sum_k q_k$, so $\Phi_w^F = \Phi_w$, and every connection carries exactly the reference rate:

$$q^F = (2.09, 3.82, 2.09) \text{ m}^3/\text{d} \quad (\Phi_w^F = 0.0278 \text{ bar}).$$

The legacy factors are therefore exact at the state they were computed for, provided Flow assigns each connection the same potential as the fracture code does.

4.3 A depth-dependent mismatch: crossflow

Take a mismatch that is linear in depth,

$$s_k = -\gamma(z_k - z_o), \quad \gamma = 0.004 \text{ bar/m} (= 0.4 \text{ bar per 100 m}), \quad (14)$$

that is, $s = (+0.040, 0, -0.040)$ bar. This is less than 5% of the hydrostatic gradient, and corresponds for instance to a wellbore column that is about 40 kg/m³ lighter than ρ_r (Section 5). Because s is antisymmetric and $T_1 = T_3$, $\sum_k (T_k + M_k)s_k = 0$ and $\Phi_w^F = 0.0278$ bar is unchanged, but the connection rates become

completion	$\Phi_w^F + s_k$ [bar]	fracture part	matrix part	total [m ³ /d]
1 (top)	0.0678	3.13	1.97	5.10
2 (middle)	0.0278	3.01	0.81	3.82
3 (bottom)	-0.0122	-0.56	-0.36	-0.92

The bottom completion produces and the top completion takes the surplus, although the fracture model (Figure 2) injects into all layers. Figure 3 shows why. The per-connection overpressure is only $\Phi_w^F = Q / \sum_k (T_k + M_k) = 8/288.1 = 0.0278$ bar. The mismatch at the bottom completion, $\gamma(z_3 - z_o) = 0.04$ bar, is larger than that overpressure, so the bottom connection potential in Flow falls below the reservoir potential.

4.4 Critical rate

From (11) and (13) with $\Phi_k = 0$, connection j reverses when $\Phi_w^F + s_j < 0$. The first connection to reverse is the one with the smallest s_j , and it reverses below the critical rate

$$Q_{crit} = \sum_k (T_k + M_k) (s_k - s_{min}) \quad (15)$$

and for a linear mismatch $s_k = -\gamma(z_k - z_o)$, $Q_{crit} = \gamma \sum_k (T_k + M_k) (z_{max} - z_k)$. For the example, $Q_{crit} = 11.5$ m³/d (Figure 4). Three properties of (15) explain the observations:

1. Q_{crit} is proportional to the *connection factors*. The fracture factors represent leak-off over the whole fracture face, so they are large, and they grow with the fracture area. A larger fracture crossflows up to a larger rate.
2. Q_{crit} is proportional to the mismatch γ and to the height spanned by the completions (Figure 5).
3. The reversal starts at the deepest connection when Flow's connection potential rises more slowly with depth than the reservoir potential ($\gamma > 0$). In that case the pattern is always injection at the top and production at the bottom.

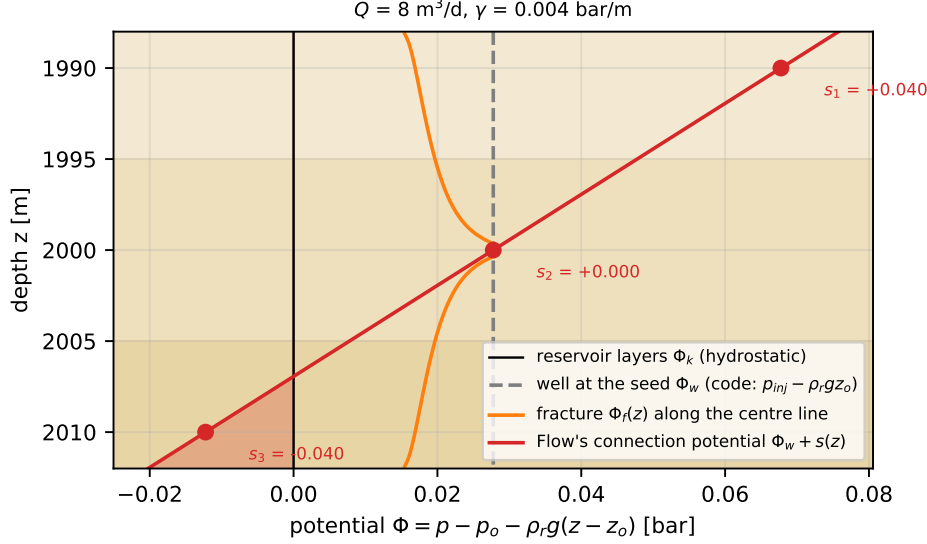


Figure 3: Potentials along the fracture height. The fracture potential (orange) is above the reservoir potential (black) everywhere, so leak-off is positive in all layers. Flow instead assigns each connection the potential $\Phi_w + s(z)$ (red). Below the depth where the red line crosses zero, the connection potential is lower than the reservoir potential, and Flow produces through the fracture CTF (shaded).

4.5 Equivalent circuits

The source of the problem is the change of network topology (Figure 6). In the fracture model, the fracture is a node with its own potential Φ_F . It is fed by the well and leaks off to the layers, and its potential is set by its own mass balance and its own fluid column. In Flow, this node is eliminated. Each layer is connected straight to the wellbore through T_k , and the potential at the end of that connection is whatever Flow's wellbore column gives at that depth. The values T_k make both networks carry the same rates at one state, but only if the potentials at the connections agree. The fracture's own column (ρ_r , no capillary pressure) is replaced by the wellbore column, so any depth-dependent difference between the two drives flow through the large fracture factors.

5 Where can s_k come from?

Table 2 lists the terms by which Flow's connection potential can differ from the potential the fracture code used. The example value $\gamma = 0.004$ bar/m stands for any of them.

In this particular example, all three completions exist in the deck with their own depths, so the seed-depth term does not apply. The fracture CTF is added to the existing connection, which keeps its deck depth. For a horizontal well, or a vertical well completed only at the seed, the fracture *creates* the upper and lower connections. Before the depth fix, those connections were placed at z_o , which gives $\gamma \approx \rho_r g$, about 25 times the example value, and a crossflow of tens of m^3/d at $Q = 8 \text{ m}^3/\text{d}$.

6 Non-hydrostatic layers and the singular legacy formula

Suppose the layer potentials genuinely differ, e.g. $\Phi = (-0.03, 0, +0.03)$ bar (deeper layer at a higher potential). Then the fracture *itself* connects the layers. When Φ_w is low, the bottom layer feeds the fracture ($q_3 < 0$). This is physical crossflow, and the fracture model computes it. The legacy formula (9) divides by $\Phi_w - \Phi_3$, which changes sign at a different well potential ($\Phi_w = \Phi_3 = 0.03$ bar) than q_3 does ($\Phi_w = 0.0407$ bar; the two differ by the pressure drop between well and fracture). Figure 7 shows the consequences:

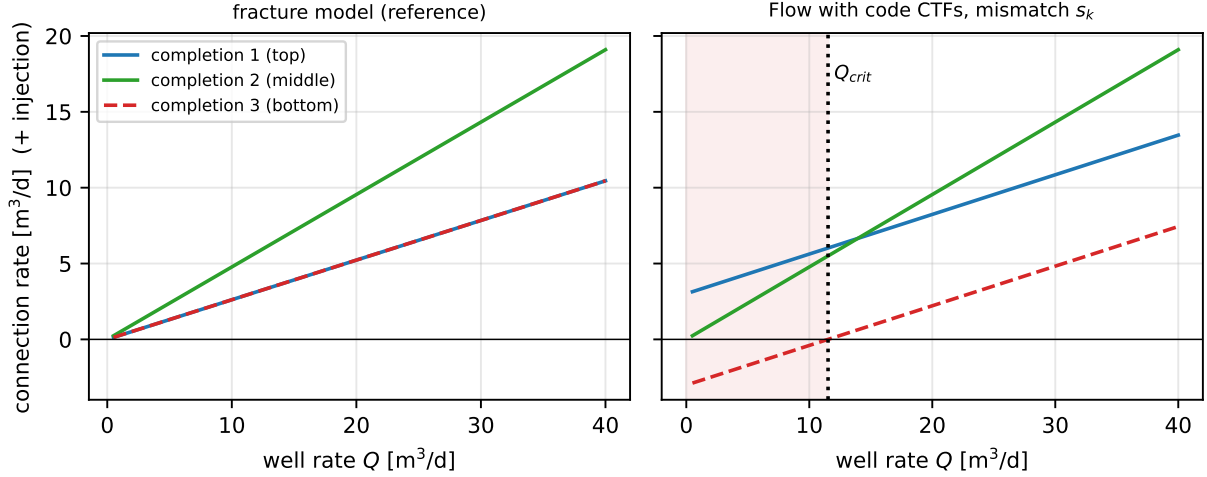


Figure 4: Connection rates as a function of the well rate. Left: fracture model (reference), top and bottom coincide. Right: Flow with the code's CTFs (fixed point of the sequential coupling, legacy) and the mismatch $\gamma = 0.004$ bar/m. Below $Q_{crit} \approx 11.7$ m³/d the bottom completion produces.

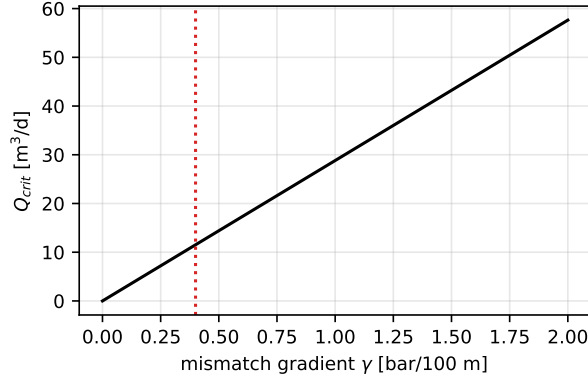


Figure 5: Critical rate (15) as a function of the mismatch gradient.

- Between the two zeros, $q_3 < 0$ but $\Phi_w - \Phi_3 > 0$. The factor is negative and is set to zero, so Flow loses the physical inflow from the bottom layer.
- Just below $\Phi_w = \Phi_3$, both are negative, and $T_3 = q_3/(\Phi_w - \Phi_3) \rightarrow +\infty$. A CTF many times L_3 is handed to Flow. Flow then reproduces q_3 only if its own $\Phi_w^F + s_3 - \Phi_3$ equals the value used in the formula. Any lag, damping or mismatch is multiplied by this factor, and Flow's rate control pins Φ_w^F near Φ_3 , with large injection at the top and production at the bottom.

This is the regime of low rates (Φ_w close to the layer potentials) in an almost hydrostatic reservoir.

7 The sequential coupling in time

To see that the reversed connection persists, the layers are given storage. Each layer is a tank with storage $C = 8$ m³/bar and support $A_q = 1000$ m³/(d bar) from the rest of the layer (an almost hydrostatic, well supported reservoir):

$$C \frac{\Phi_k^{n+1} - \Phi_k^n}{\Delta t} = -A_q \Phi_k^{n+1} + q_k^{F,n+1}, \quad \Delta t = 1 \text{ d.} \quad (16)$$

Each step reproduces the code's sequence: a fracture solve at Flow's current seed potential and layer potentials; CTFs from (9), damped as in `wellIndicesAvg`; and an implicit Flow step with

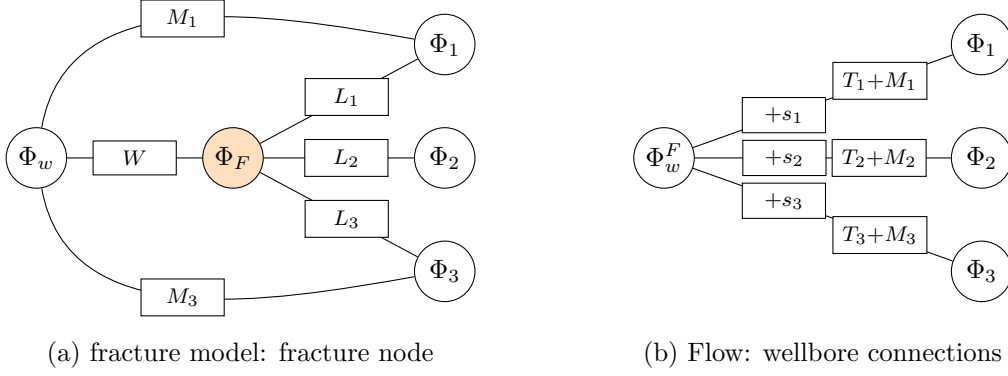


Figure 6: (a) The fracture model: the fracture is a node with its own potential Φ_F and its own fluid column. (b) Flow: the fracture node is removed, and each layer sees the wellbore potential plus Flow’s column term s_k through the large factor T_k . (M_2 is omitted from (a) for clarity.)

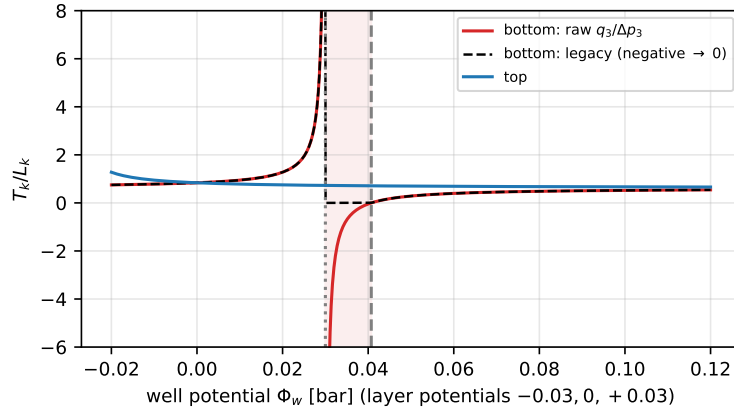


Figure 7: Legacy factor T_k/L_k as a function of the well potential, with layer potentials $(-0.03, 0, +0.03)$ bar. Dotted: $\Phi_w = \Phi_3$ (pole); dashed: $q_3 = 0$. Shaded: negative factor, set to zero.

these frozen CTFs under rate control, with s_k as above. The reference solves the fracture node implicitly with the layers and has no s_k . Figure 8 shows the result. With the legacy factors, the bottom completion produces throughout (-0.64 m³/d at the end, against $+2.16$ m³/d in the reference), and the top completion takes 4.91 m³/d.

8 Remedies evaluated on the example

Option 2 (wi_flow_column). The CTF is back-calculated against Flow’s own connection potential:

$$T_k = \frac{q_k}{\Phi_w^F + s_k - \Phi_k}. \quad (17)$$

At the bottom completion $q_3 > 0$ but $\Phi_w^F + s_3 < 0$, so the factor is negative and is set to zero. The fracture stops feeding the bottom layer in Flow, which is not what the fracture model says, and the bottom completion now only crossflows through its matrix CTF (-0.20 m³/d). The rates are $(3.65, 4.55, -0.20)$ m³/d. Option 2 removes the *amplification* by the fracture factor, but it inherits the singularity of Section 6, because it still divides by a potential difference that can vanish at low rates.

source	s_k	status / magnitude
Fracture connection created at the seed depth (<code>perf.depth = origo_[2]</code>), standard well	$-\rho_r g(z_k - z_o)$, i.e. $\gamma \approx 0.1$ bar/m	Fixed in commit <code>c9d4b6e</code> (depth = cell depth). Only for connections the fracture <i>creates</i> ; deck completions keep their depth. In this example $Q_{crit} = 288$ m ³ /d, and at $Q = 8$ the rates are (77.4, 3.8, -73.2) m ³ /d.
Wellbore column density ρ_{wb} (Flow) $\neq \rho_r$ (<code>dens.cells</code> , CTF)	$(\rho_{wb} - \rho_r) g(z_k - z_o)$	For water injection, Flow uses $\rho_{w,s}/B_w$ at the cell temperature and salinity, so it is nearly equal to ρ_r . Large as soon as another phase enters the wellbore (gas or CO ₂ injection, or crossflowing oil or gas), which is a positive feedback. $\gamma = 0.004$ for $\Delta\rho \approx 41$ kg/m ³ .
Cell pressure in the well model (oil or gas) vs. water pressure in the fracture code	$-(P_{c,k} - P_{c,o})$	Zero below the contacts; depth-dependent in transition zones. Option 2 removes it.
Lag and damping of the CTFs (<code>wellIndicesAvrg: (new+2*old)/3</code>), reservoir change during the step	Φ_k seen by the CTF $\neq \Phi_k$ in Flow	Not depth-linear by itself, but it is amplified by the singular formula (9) (Section 6).
Real, non-hydrostatic layer potentials	(not a mismatch)	Physical crossflow through the fracture; the fracture model shows it too (Section 6).

Table 2: Contributions to the mismatch s_k in (11).

Option 3 (`wi_fracture_pressure`). The fracture part of each connection is driven by the fracture’s own potential, shifted with the well:

$$q_k^F = L_k(\Phi_w^F + (\bar{\Phi}_{F,k} - \Phi_w) - \Phi_k) + M_k(\Phi_w^F + s_k - \Phi_k). \quad (18)$$

The mismatch s_k no longer enters the fracture part. The fracture injects into every layer, as in the fracture model (+1.28 m³/d into the bottom layer), and only the ordinary matrix completion feels Flow’s column (-0.36 m³/d). That matrix crossflow is Flow’s normal well behaviour and would occur without a fracture. The rates are (3.26, 3.82, 0.93) m³/d, and in time the bottom completion stays at +1.10 m³/d (Figure 8).

9 The conductivity mode (`wi_upscaling = conductivity`)

9.1 What the code does

In `conductivity` mode, `Fracture::wellIndices()` does not divide by a potential difference. The CTF is the summed leak-off conductance itself, optionally followed by a sign gate and by one global normalisation factor:

```
ctf = trans_cells[i]; // sum of leakof_/mobility
const bool not_fed = (q < 0) || (dp_well <= 0 && q > 0);
if (wi_sign_gate && not_fed) ctf = 0; // default: wi_sign_gate = false
sum_q += q_cells[i];
sum_ctf_dp += ctf * mob * std::max(dp_well, wi_dp_floor); // floor: 1e4 Pa
...
alpha = std::min(sum_q / sum_ctf_dp, wi_alpha_max); // if wi_flux_normalization (default on)
perf.ctf *= alpha; // and both sums are positive
```

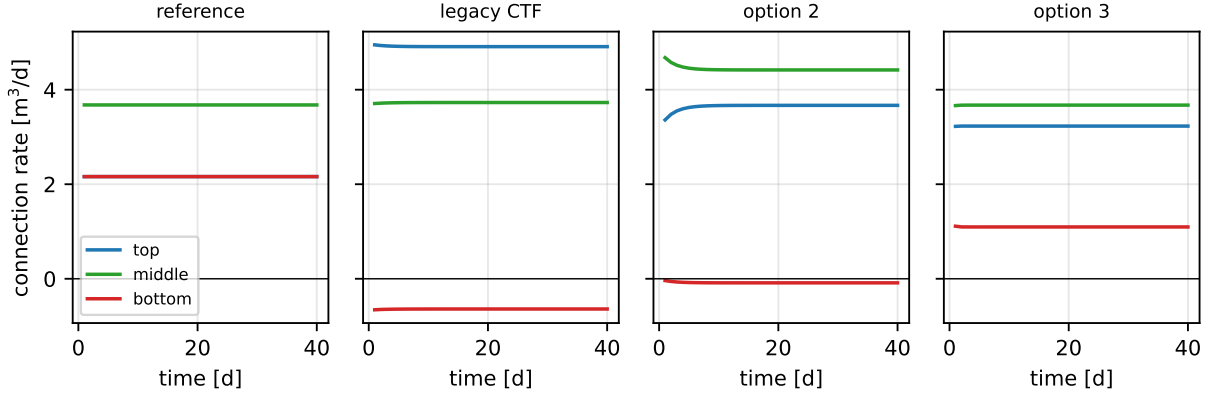


Figure 8: Connection rates over 40 days of sequential coupling at $Q = 8 \text{ m}^3/\text{d}$, $\gamma = 0.004 \text{ bar/m}$.

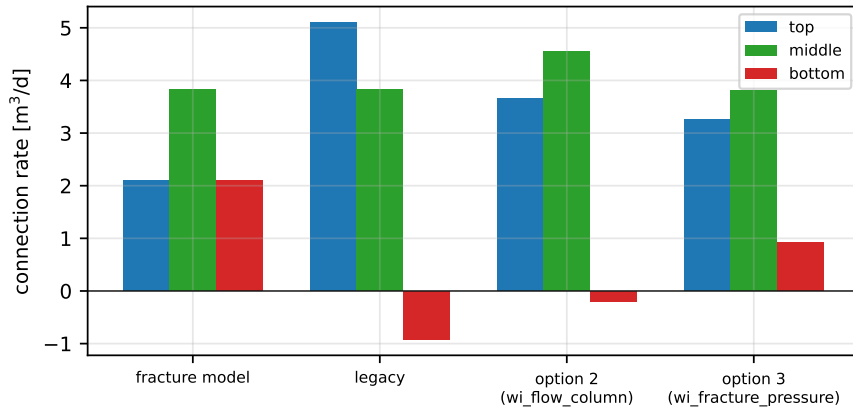


Figure 9: Connection rates at $Q = 8 \text{ m}^3/\text{d}$ and $\gamma = 0.004 \text{ bar/m}$ (fixed point of the sequential coupling, frozen layer potentials).

In potentials, with $p_{fl} = 0.10 \text{ bar}$ (`wi_pressure_floor`) and $\alpha_{\max} = 2$,

$$T_k = \alpha L_k \quad (0 \text{ if gated}), \quad \alpha = \min\left(\frac{\sum_k q_k}{\sum_k L_k \max(\Phi_w - \Phi_k, p_{fl})}, \alpha_{\max}\right). \quad (19)$$

Here $\Phi_w - \Phi_k$ is the code's `dp_well`: the fracture-side potential difference, or Flow's own difference when `wi_flow_column` is set. The factor is positive by construction, so the singularity of Section 6 cannot occur. The crossflow mechanism of (11) is unchanged, however: Flow still applies T_k with its own connection potential $\Phi_w^F + s_k$, and (15) still holds with $T_k = \alpha L_k$. What changes is the size of T_k , and therefore Q_{crit} .

9.2 Three regimes

Without normalisation (`wi_flux_normalization = false`). Then $T_k = L_k$. This is larger than the legacy factor ($T_k^{\text{leg}}/L_k \approx 0.62\text{--}0.68$ here), because L_k ignores the pressure drop between the well and the fracture. Two consequences follow. Even without a mismatch, Flow injects 1.54 times the modelled leak-off through the fracture, at a lower well potential ($\Phi_w^F = 0.0202$ instead of 0.0278 bar). And the critical rate grows to

$$Q_{crit} = \sum_k (L_k + M_k)(s_k - s_{\min}) = 15.8 \text{ m}^3/\text{d} \quad (\text{legacy: } 11.5).$$

At $Q = 8 \text{ m}^3/\text{d}$, the bottom completion produces $-2.06 \text{ m}^3/\text{d}$.

Normalisation without floor ($p_{fl} \rightarrow 0$). Here $\alpha = \sum_k q_k / (\sum_k L_k \Phi_w) = \sum_k T_k^{\text{leg}} / \sum_k L_k = 0.651$. This is the legacy factor averaged over the connections and applied uniformly, so the result is practically the legacy one ($Q_{crit} \approx 11.7 \text{ m}^3/\text{d}$, bottom $-0.95 \text{ m}^3/\text{d}$).

Defaults (normalisation with $p_{fl} = 0.10 \text{ bar}$). At low rates $\Phi_w < p_{fl}$, and the floor takes over the denominator. With $\sum_k q_k = g \Phi_w$, where $g = 200.8 \text{ m}^3/(\text{d bar})$ is the fracture model's injectivity,

$$\alpha = \frac{g}{\sum_k L_k} \frac{\Phi_w}{p_{fl}}, \quad \frac{\text{fracture injection in Flow}}{\text{modelled leak-off}} = \frac{\alpha \sum_k L_k \Phi_w}{g \Phi_w} = \frac{\Phi_w}{p_{fl}}. \quad (20)$$

The normalisation was meant to preserve the total exchange rate. Below the floor it instead *throttles* the fracture connections in proportion to Φ_w/p_{fl} . Flow then needs a higher well potential to place the rate. The coupled fixed point satisfies

$$Q = \frac{g}{p_{fl}} \Phi_w^2 + \sum_k M_k \Phi_w \quad \implies \quad \Phi_w = 0.0450 \text{ bar at } Q = 8 \text{ m}^3/\text{d}, \quad \alpha = 0.293, \quad (21)$$

so Flow injects only 0.45 of the modelled leak-off through the fracture. Crossflow starts when Φ_w falls to $-s_{\min}$, which gives

$$Q_{crit} = \frac{g}{p_{fl}} s_{\min}^2 + \left(\sum_k M_k \right) |s_{\min}| = 6.7 \text{ m}^3/\text{d}. \quad (22)$$

This is *quadratic* in the mismatch. The floor therefore helps against small mismatches ($|s_{\min}| < p_{fl}$), and at $Q = 8$ all completions inject (4.34, 3.40, 0.26 m^3/d). It gives no protection against large ones: for the seed-depth mismatch, Φ_w exceeds p_{fl} , α returns to $g/\sum_k L_k$, and Q_{crit} is the same $\approx 288 \text{ m}^3/\text{d}$ as in legacy mode. Note that the crossflow is suppressed by misrepresenting the fracture: the BHP in Flow is higher, and the fracture carries less, than the fracture model says.

Sign gate. With the fracture-side `dp_well`, the gate never acts in the mismatch case: the fracture model sees positive leak-off and a positive potential difference in every layer. Combined with `wi_flow_column`, the gate removes the fracture part of every connection where Flow's own potential difference is not positive. The fracture part then cannot crossflow. The remaining production below $Q \approx 7.7 \text{ m}^3/\text{d}$ in Figure 10 is the matrix completion alone.

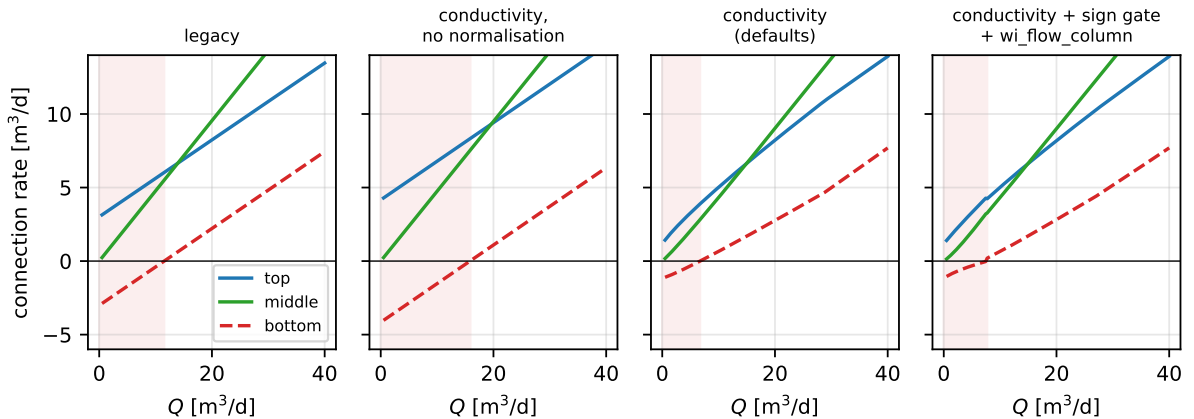


Figure 10: Connection rates versus well rate for $\gamma = 0.004 \text{ bar/m}$ (fixed point of the sequential coupling). Shaded: bottom completion produces.

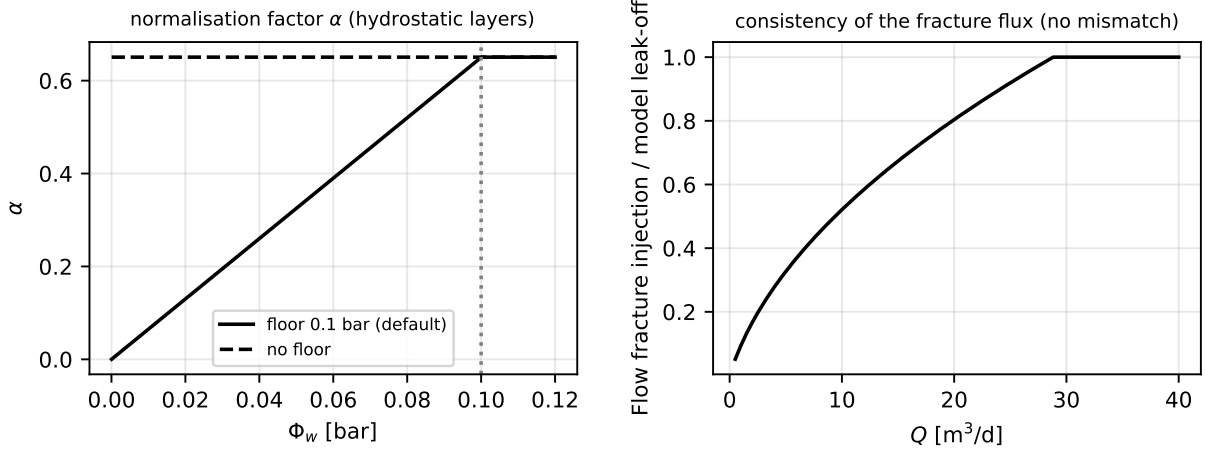


Figure 11: Left: normalisation factor α versus well potential, hydrostatic layers. Below the floor (dotted), α falls linearly to zero. Right: fracture injection in Flow relative to the modelled leak-off at the coupled fixed point, without mismatch. It is Φ_w/p_{fl} at low rates, see (20).

mode	α	Φ_w^F [bar]	top	middle [m ³ /d]	bottom	Q_{crit} [m ³ /d]
fracture model (reference)	–	0.0278	2.09	3.82	2.09	–
legacy	–	0.0278	5.10	3.82	-0.92	11.5
conductivity, no normalisation	1.000	0.0202	6.26	3.80	-2.06	15.9
conductivity, no floor	0.651	0.0278	5.27	3.68	-0.95	11.7
conductivity, defaults	0.293	0.0450	4.34	3.40	0.26	6.7
conductivity + gate + wi_flow_column	0.293	0.0450	4.34	3.40	0.26	7.7

Table 3: Fixed points of the sequential coupling with mismatch $\gamma = 0.004$ bar/m. Q_{crit} is read from the rate sweeps.

9.3 Comparison at $Q = 8$ m³/d

9.4 Non-hydrostatic layers and dynamics

With genuinely different layer potentials (Section 6), the conductivity factor has no pole (Figure 12). All connections share the same α , so the bottom factor equals the top one. Two artefacts remain. First, α jumps to 1 when $\sum_k q_k \leq 0$, because the normalisation is skipped. Second, the sign gate removes the physical inflow from the bottom layer while $q_3 < 0$.

In the time-dependent coupling (Figure 13), the bottom completion ends at -1.59 m³/d without normalisation and at $+0.38$ m³/d with the defaults (reference $+2.16$, legacy -0.64).

10 Formation damage and the m51w deck

10.1 Properties from test/m51w/TEST.DATA

The geometry is now taken from the deck (Figure 14). The cells are $2 \times 2 \times 2$ m. The injector B-3H is completed in K=36–40 (centres 1971–1979 m), and the seed is at K=38 ($z_o = 1975$ m). A circular fracture of radius $R = 3$ m again covers three completions (K=37–39); K=36 and K=40 are matrix-only. The fracture settings are those of `params/seq_implicit_fb_robust_v2_test3.json`.

Why the fracture is open at a small overpressure. At the seed, the total horizontal stress from STREQUIL is $\sigma_h \approx 330$ bar, against a pore pressure of 192.5 bar. Water is injected at 30 °C into a 90 °C reservoir (WTEMP, RTEMPVD). The thermal stress reduction of fully cooled rock,

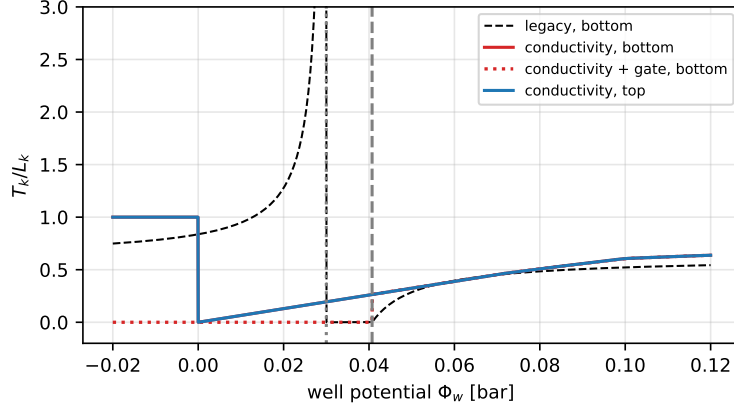


Figure 12: Conductivity factors $T_k/L_k = \alpha$ with layer potentials $(-0.03, 0, +0.03)$ bar, compared with the legacy bottom factor.

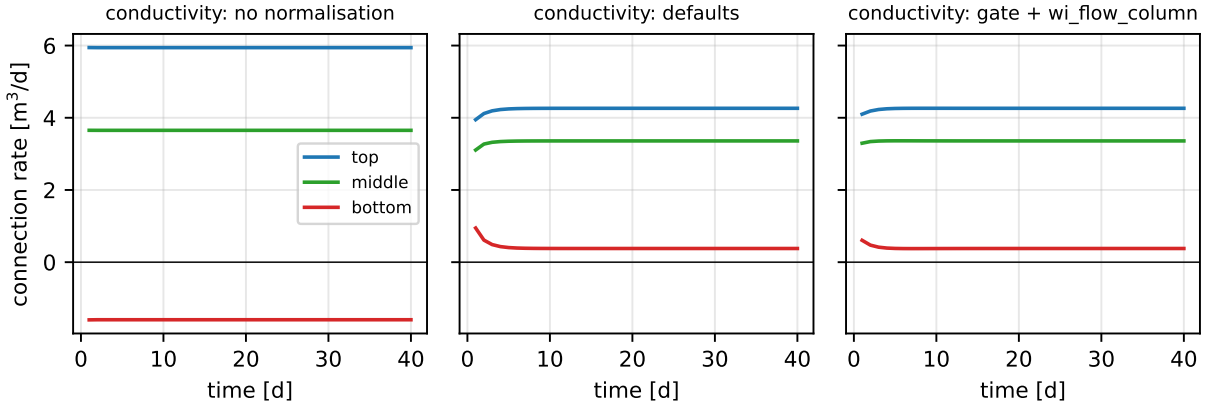


Figure 13: Sequential coupling over 40 days, conductivity mode, $Q = 8 \text{ m}^3/\text{d}$, $\gamma = 0.004 \text{ bar/m}$.

$E\alpha_T\Delta T/(1 - \nu) \approx 388 \text{ bar}$, exceeds the effective stress. The cooled region therefore fractures at a pressure close to the reservoir pressure, which is the situation assumed throughout this note.

10.2 The margin in m51w

The fracture leak-off conductance is $\sum_k L_k = 2422 \text{ m}^3/(\text{d bar})$ for only 28 m^2 of fracture: $85.6 \text{ m}^3/(\text{d bar})$ per m^2 at 1000 mD and $d = 0.5 \text{ m}$. At the deck rate, the overpressure per connection relative to the adjacent cells is $\Phi_w = 0.0315 \text{ bar}$. That is about $1/6$ of the hydrostatic difference across a single 2 m cell (0.197 bar). With frozen layer potentials, the critical mismatch at the deck rate is only $\gamma = 0.0080 \text{ bar/m}$ for legacy (about 8.1% of the hydrostatic gradient) and 0.0150 bar/m for conductivity (Figure 15).

The layers re-equilibrate. The frozen-layer picture does not hold over Flow's time steps. With $k = 1000 \text{ mD}$, the pressure diffusion time over 10 m is of the order of seconds. Each layer's pressure therefore follows its own injection, and a layer that receives less water stays at a lower pressure, which restores its drive. We represent each completed layer by a tank with storage $C = 2 \text{ m}^3/\text{bar}$, drainage $A_q = 110 \text{ m}^3/(\text{d bar})$ to the far field, and vertical coupling $T_v = 430 \text{ m}^3/(\text{d bar})$ to the neighbouring layers. These are order-of-magnitude estimates from the deck: drainage to the free bottom boundary, and $k_v = 100 \text{ mD}$ over a $20 \times 20 \text{ m}^2$ near-well area. The layers then rise by about 0.2 bar above their initial potential. Sustained crossflow needs a mismatch large enough to

quantity	value	source
permeability (completed layers), k_v/k_h	1000 mD, 0.1	EQUALS PERMX, MULTIPLY PERMZ
water density, viscosity, B_w (195 bar)	1004 kg/m ³ , 0.398 cP, 1.0219	DENSITY, PVTW, EQUIL
mobility (water zone, WOC above the model)	$\lambda = 1/\mu_w$	EQUIL WOC 1000 m
rate	100 sm ³ /d = 102.2 m ³ /d	WCONINJE
wellbore radius, Peaceman radius, $\ln(r_o/r_w)$	0.108, 0.396 m, 1.299	COMPDAT diameter 0.216
matrix CTF per completion	$M_k = 207 \text{ m}^3/(\text{d bar})$	Peaceman, $h = 2 \text{ m}$
leak-off distance, faces	$d = 0.5 \text{ m}$, $n_s = 2$	calculate.dist, legacy model
leak-off conductance per m ² of fracture	85.6 m ³ /(d bar)	$n_s \lambda k/d$
E , ν	14 GPa, 0.35	YMODULE, PRATIO
net pressure (example choice), $w_{\max}$	20 bar, 0.96 mm	Sneddon
feed	cells within 1 m of the well	well_source_all_perfs, radius 1.0
filter cake	$k_c = 10 \text{ mD}$, $\phi_c = 0.3$, LINRAD	WINJDAM
particle concentration	10 ppm	WINJFCNC

Table 4: Parameters taken from the m51w deck and its fracture settings.

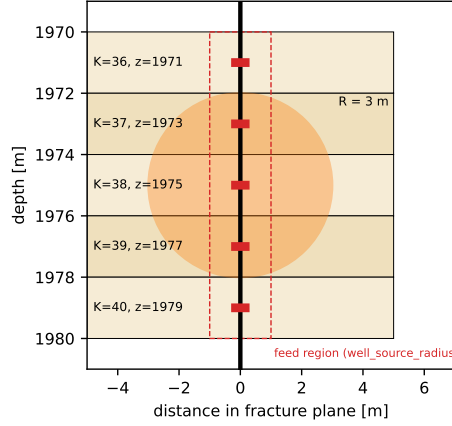


Figure 14: m51w geometry: five completions in 2 m cells, a fracture of radius 3 m around the seed (K=38), and the feed region of `well_source_all_perfs`.

overcome this build-up, which gives the quasi-steady critical mismatch

$$\gamma_{crit} = 0.024 \text{ (legacy)}, \quad 0.029 \text{ (conductivity)}, \quad 0.020 \text{ bar/m (option 3)},$$

i.e. 20–30 % of the hydrostatic gradient. Below that, a mismatch only skews the distribution. At $\gamma = 0.01 \text{ bar/m}$, for example, the top completion takes 14.9 m³/d and the bottom one 6.2 m³/d (legacy), against 10.5 each in the reference. Mismatches of 20–30 % of the hydrostatic gradient come from the seed-depth placement of connections the fracture creates (before the depth fix), or from gas in the wellbore column. They do not come from the small density or capillary differences of a water-filled deck. The physical drainage to the free bottom boundary (BCPROP 6 FREE) gives a potential gradient of $\approx 0.0046 \text{ bar/m}$ that *decreases* with depth, which favours injection at the bottom.

With option 3, which is what the m51w settings use (`wi_fracture_pressure = true`), the completion that reverses first is K=40, the matrix-only completion below the fracture. The fracture part is immune to s_k , but it pressurises the fractured layers, and through the vertical coupling also K=40, while K=40’s matrix connection still sees Flow’s column.

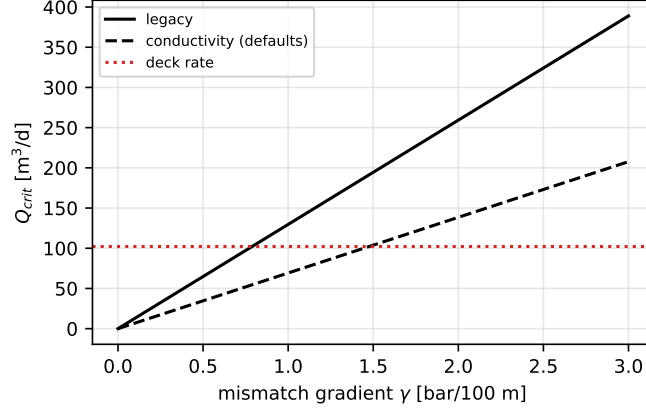


Figure 15: m51w, frozen layer potentials: critical rate (15) against the mismatch gradient. The deck rate is crossed at $\gamma = 0.0080$ (legacy) and 0.0150 bar/m (conductivity).

Layer pressurisation and the legacy formula. For a mismatch above the threshold, $\gamma = 0.030$ bar/m in the dynamics below, the two effects reinforce each other in legacy mode. Flow injects more into the top fractured layer ($K=37$), which raises that layer’s potential above the well potential ($\Phi_{37} = 0.235$ vs. $\Phi_w = 0.223$ bar). The fracture model, which is solved with Flow’s layer potentials, then sees *inflow* from $K=37$ ($q = -8.6$ m³/d). The legacy formula divides a negative leak-off by a negative potential difference and returns a large positive factor, so Flow keeps injecting 33.7 m³/d there. $K=39$ is the opposite case: the model leaks off 48.1 m³/d there, but Flow injects only 8.3 m³/d (Figure 18, left). This is the singular regime of Section 6, reached through the layers’ own response.

10.3 Formation damage (WINJDAM)

Two cakes grow independently in the code.

Matrix completions (`WellFilterCake`, `LINRAD`). The matrix share of the injected water of each connection deposits particles on the perforation area $A_c = 2\pi r_w L$:

$$\Delta h = \frac{q_k^{mat} c \chi \Delta t}{A_c(1 - \phi_c)}, \quad S_k += \frac{K}{k_c} \ln \frac{r_w + h + \Delta h}{r_w + h}, \quad M_k \leftarrow M_k \frac{D}{D + S_k}, \quad (23)$$

with $K/k_c = 100$, $D = \ln(r_o/r_w) = 1.299$, and the crossflow factor $\chi = \sum q / \sum q^+$. The multiplier is applied to the matrix part of T_w only (`WellInterface::getTw`). Because D is small (2 m cells) and $K/k_c = 100$, the damage is fast: at the seed completion the multiplier is 0.76 after 10 days and 0.29 after 100 days (Figure 16).

Fracture faces (`Fracture::updateFilterCakeProps`, `updateLeakoff`). Each fracture cell accumulates a cake from its *modelled* positive leak-off, scaled by Flow’s fracture rate (`scale_filttrate`). The cake acts in series with the reservoir path on each face:

$$\Delta h_{c,i} = \frac{\max(\ell_i(\Phi_i - \Phi_{k(i)}), 0) f_s c \Delta t}{\Delta A_i(1 - \phi_c)}, \quad \ell_i = n_s / \left(\frac{d}{\lambda k \Delta A_i} + \frac{h_{c,i}/n_s}{\lambda k_c \Delta A_i} \right). \quad (24)$$

The fracture face area is much larger, so this cake grows slowly: the leak-off conductance is at 0.75 of its clean value after 100 days at the seed layer.

10.4 Effects of the damage on crossflow

Figure 17 compares 100 days of sequential coupling with and without `WINJDAM`, at $\gamma = 0.030$ bar/m.

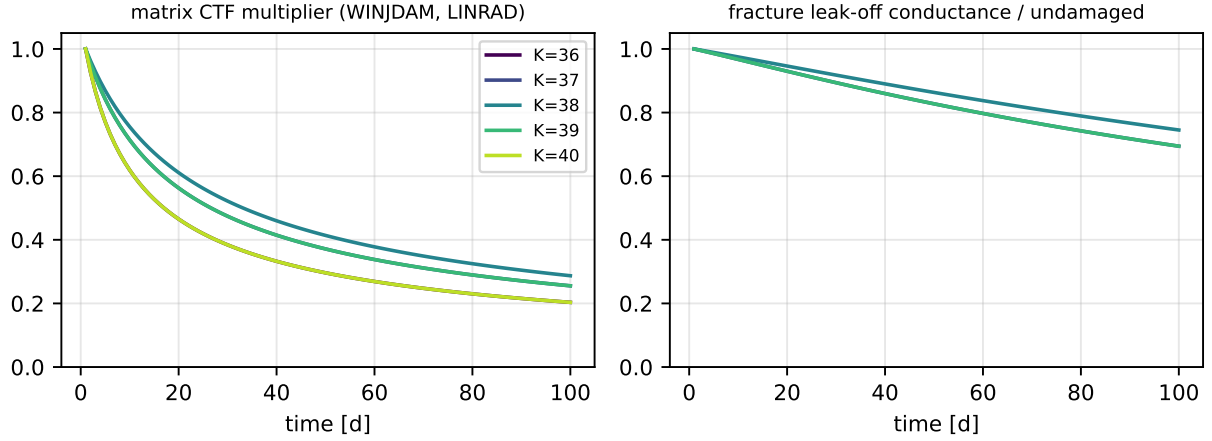


Figure 16: Reference run with WINJDAM: matrix CTF multipliers (left) and fracture leak-off conductance relative to the clean fracture (right).

1. **The well pressure rises.** The conductances fall, so the BHP rises to place the rate (Φ_w : $0.224 \rightarrow 0.255$ bar in the reference), and every connection gets more margin against the mismatch. The quasi-steady critical mismatch at the day-100 damage state increases to 0.031 (legacy), 0.035 (conductivity) and 0.026 bar/m (option 3).
2. **Only injecting completions are damaged.** The cake grows only where water goes in. A crossflowing completion stays clean (K=40 keeps a multiplier of 1.00 in the legacy run), while the injecting ones lose 60–85 % of their matrix CTF. The net effect is still less crossflow, because the rising BHP dominates: at K=40 the rate goes from -2.51 to -0.49 m³/d (legacy) and from -5.43 to -1.87 m³/d (option 3). In conductivity mode it changes sign, from -0.54 to $+1.66$ m³/d.
3. **Injection moves into the fracture.** The matrix-only completions K=36 and K=40 drop from 10.5 to 3.85 m³/d in the reference, and the fracture takes the rest.
4. **The fracture cake goes where the model leaks off, not where Flow injects.** In the legacy run, the model sees inflow from K=37 ($q = -0.9$ m³/d), so no cake forms there (leak-off conductance 1.00 of clean), while Flow injects 57.3 m³/d through it. K=39, where the model leaks off, is damaged to 0.52 of clean. The damage thus pushes even more of Flow’s injection to the top: $33.7 \rightarrow 57.3$ m³/d (Figure 18, right). This is a positive feedback on the skew, built on the inconsistency between Flow’s connection rates and the fracture model’s leak-off. With option 3 the two are consistent, and the fracture cake is evenly distributed.

10.5 What this means for m51w

- The overpressure per connection is tiny (0.03 bar), but the layers’ own build-up protects against small mismatches. Sustained crossflow needs a mismatch of 20–30 % of the hydrostatic gradient. Such mismatches came from the seed-depth placement of new connections before the depth fix. They can also occur when the fracture grows beyond K=36–40, so that new connections are created.
- With the m51w settings (option 3), the fracture connections are immune to the mismatch. Remaining crossflow goes through the matrix-only completions outside the fracture, or is physical (non-hydrostatic layer potentials).

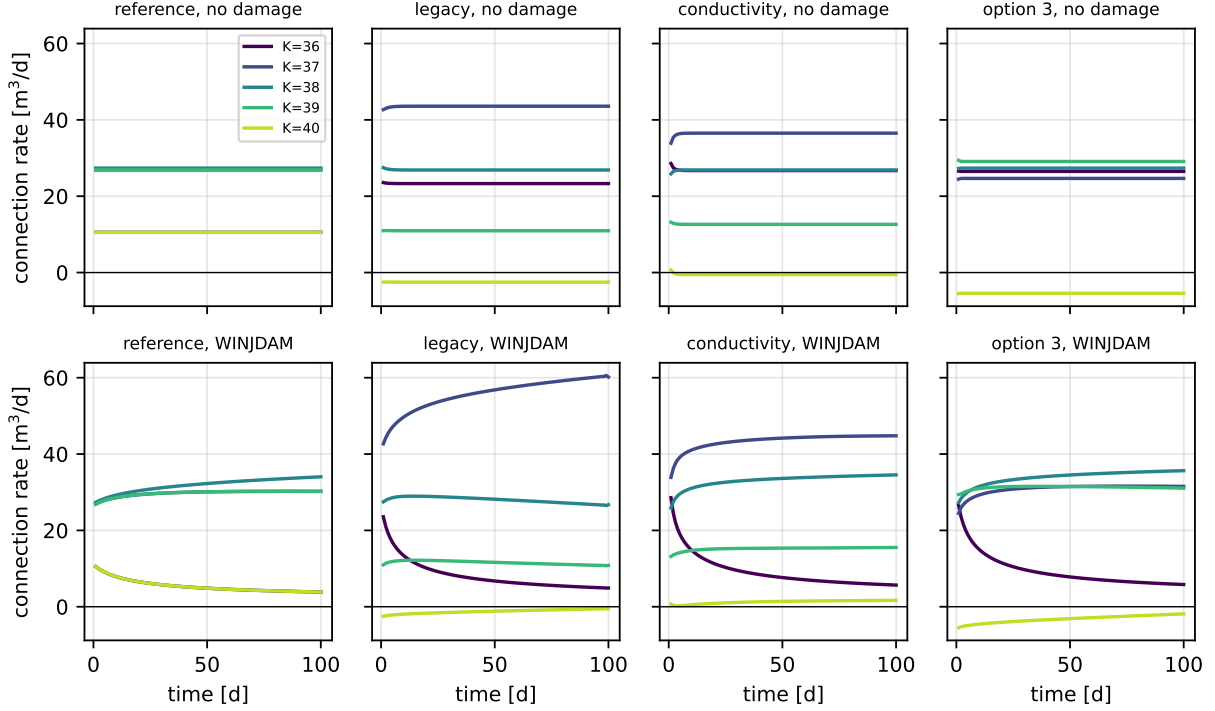


Figure 17: m51w properties, sequential coupling over 100 days at the deck rate with $\gamma = 0.030$ bar/m. Top: no damage. Bottom: WINJDAM/WINJFCNC as in the deck. Total connection rates per completion.

- WINJDAM reduces crossflow mainly by raising the BHP. In legacy and conductivity modes it can also reinforce a skew, because the fracture cake follows the fracture model's leak-off rather than Flow's connection rates.
- Model limitations: tanks instead of a grid, a prescribed fracture size and width, and no thermal effects on the fluids (none are in the deck). The numbers indicate orders of magnitude and directions, not a prediction of the m51w run.

11 Summary

1. The legacy CTF $T_k = q_k/(\Phi_w - \Phi_k)$ is exact only at the state it was computed for, and only if Flow gives every connection the potential Φ_w that the formula assumed.
2. Flow applies the factors with its own wellbore column and cell pressures. Any depth-dependent difference s_k acts as an extra driving potential on the fracture factors. Under rate control, the deepest connection reverses for $Q < Q_{crit} = \sum_k (T_k + M_k)(s_k - s_{min})$.
3. The fracture factors are large (leak-off over the whole face), while the overpressure per connection at low rates is $Q/\sum_k (T_k + M_k)$, a few hundredths of a bar. A mismatch of the same order (here ± 0.04 bar over ± 10 m) is enough to produce injection at the top and production at the bottom.
4. Without the depth fix, connections created by the fracture were placed at the seed depth ($\gamma \approx \rho_r g$), which gives Q_{crit} of hundreds of m^3/d for this small fracture.
5. When the layer potentials truly differ, the fracture model also crossflows (physical). The legacy formula is singular there, and it zeroes or inflates the bottom factor.
6. `wi_upscaling = conductivity` removes the pole but not the mechanism. Without normalisation, $T_k = L_k$ is larger than the legacy factor, so it crossflows more ($Q_{crit} = 15.9$

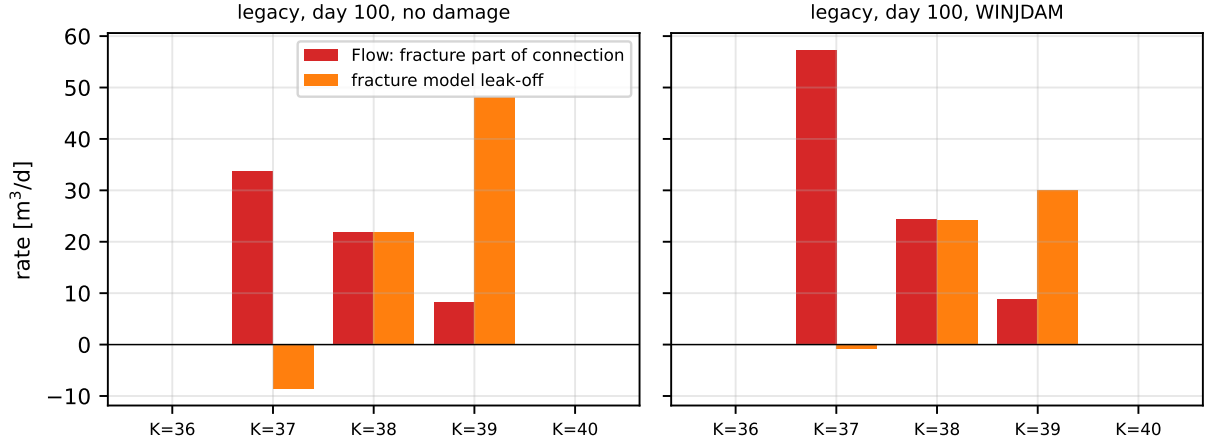


Figure 18: Legacy mode, day 100: fracture part of Flow’s connection rates against the fracture model’s own leak-off per completion. The two disagree in sign at K=37, and the cake is laid down according to the model.

vs. $11.5 \text{ m}^3/\text{d}$). With the default normalisation, the pressure floor scales the fracture connections by Φ_w/p_{fl} at low rates. This lowers Q_{crit} ($6.7 \text{ m}^3/\text{d}$) only by making Flow inject less through the fracture than the fracture model computes, and it does not help against mismatches larger than the floor.

7. Driving the fracture part with the fracture’s own pressure (option 3) removes the fracture-induced crossflow. Option 2 removes the mismatch but keeps the singular division. The structural alternative is the embedded coupling, where the fracture node stays in the flow problem.
8. With the m51w properties, the layers’ own pressure build-up raises the threshold for sustained crossflow to a mismatch of 20–30 % of the hydrostatic gradient. WINJDAM raises it further through a higher BHP. In legacy and conductivity modes, however, the fracture cake is deposited according to the fracture model’s leak-off, not Flow’s connection rates, which can reinforce an existing skew.

A Reproducing the numbers

All numbers and figures are generated by `crossflow_example.py` and `m51w_case.py` in this directory (run both, then `pdflatex crossflow_example.tex`). The two-dimensional fracture solve uses `scipy.sparse`; the response to the well and layer potentials is computed once and then used for every rate, coupling iteration and time step.